Shoshana McDermott

(504) 406-5124 | smcdermott@hmc.edu | github.com/smcdermott-commits | linkedin.com/in/shoshana-mcd

Education

Harvey Mudd College Aug. 2026 – May 2030
Bachelor of Science in Engineering, National Merit Scholar *Claremont, CA*

Benjamin Franklin High School Aug 2022 – May 2026
Summa Cum Laude, GPA 4.39, National Honor Society, Honor Roll, 14 AP Courses *New Orleans, LA*

Experience

Mudd Amateur Rocketry Club (MARC) Sep 2026 – Present
Propulsion Team Member *Claremont, CA*

- Model active-guidance fin tabs in SolidWorks for computational fluid dynamics simulations, with results feeding the avionics team.
- Help manufacture an L2 rocket, launching mid-November 2026, to test active guidance.
- Build a personal Level 1 high-power rocket, with certification planned for November.

Harvey Mudd Makerspace Oct 2026 – Present
Database Steward *Claremont, CA*

- Audit supply records to keep inventory information accurate and up to date.
- Work with the inventory team to plan stock so users have reliable access to materials, balanced against space and cost.

Mathnasium May 2026 – Aug 2026
Instructor *New Orleans, LA*

- Delivered differentiated K–12 math instruction, basic numeracy through precalculus, to 2–4 students at a time, across a student body of 50+.
- Prepared per-session progress reports to document individual student mastery and progress.

Physics in Motion Aug 2023 – May 2026
Tutor and Leadership Board Member *Benjamin Franklin High School*

- Managed a team of 30+ tutors, delegated worksheet creation, managed onboarding responsibilities, and coordinated tutoring schedules.
- Built an automated Google Sheets tracking system for tutor session and worksheet data, reducing time to compile required reporting forms to under a minute.

Projects

University of New Orleans NASA USLI Design Team Member July – Aug 2025

- Collaborated with a UNO engineering team to design and build a high-powered rocket for NASA's University Student Launch Initiative.
- Modeled a 3D-printable avionics bay holder in SolidWorks to fit component constraints and secure onboard components.
- Simulated flight performance in OpenRocket to assess design effects on rocket stability and validate structural decisions.

Wally Wiki (wally-wiki.vercel.app) Sep 2026 – Oct 2026

- Designed and deployed a resource directory for Harvey Mudd students with admin-gated content management.
- Integrated Gemini and Groq with automatic failover and retry logic to auto-populate resource details from submitted URLs.
- Reduced initial page load time by ∼40% by parallelizing network requests and trimming database queries to required fields.

Skills

Engineering Software: SolidWorks, OnShape, OpenRocket, LaTeX
Programming & Data: Python, Microsoft Excel, Google Sheets
Spoken Languages: French (fluent, DELF B2), Spanish (conversational)